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26 July 1967
Superseding
XWS 8509
12 May 1967

MILITARY SPECIFICATION

PACK ASSEMBLY, PARACHUTE

MARK 33 MOD 0

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 This specification establishes the requirements of one type of Parachute Pack Assembly Mark 33 Mod 0 hereafter referred to as Pack Assembly. (See 6.1 for intended use and description.)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein except as noted.

SPECIFICATION

MILITARY

MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for

STANDARDS

MILITARY

MIL-STD-105D	Sampling Procedures and Tables for Inspections by Attributes

FCS-1375

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MIL-STD-129 Marking for Shipment and Storage

DRAWINGS

BUREAU OF NAVAL WEAPONS

LD 497745	Release, Parachute Mk 26 Mod 0
LD 497746	Pack Assembly, Parachute Mk 33 Mod 0
LD 497770	Preserved (Packing) Condition, Parachute Pack Assembly Mk 33 Mod 0
LD 497771	Shipping (Packing) Condition, Parachute Pack Assembly Mk 33 Mod 0
DWG. 1786083	Pin

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

OD 13799	Special Manufacturing Process for the Parachute Pack Assembly Mk 33 Mod 0

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. Unless otherwise specified in the contract or order, materials and treatments of materials and parts used in the manufacture of the pack assembly shall be in accordance with the requirements specified herein and with the applicable documents listed in BuWeps LD 497746.

3.1.1 The pack assembly shall be assembled in accordance with procedures specified in NAVWEPS OD 13799.

3.2 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the pack assembly shall be manufactured using the methods and procedures proposed for production. The sample shall be tested as specified in Section 4 herein and is for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract.

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3.3 Performance and product characteristics. The pack assembly shall meet the following performance and product characteristics.

3.3.1 Nonoperating. After withdrawing pin, Drawing 1786083, Release Mechanism Mk 26 Mod 0, LD 497745 shall not operate when exposed to a constant acceleration of 4g (g = unit of gravitational acceleration) plus 0g minus 1g.

3.3.2. Operating. After withdrawing pin, Drawing 178608.3, Release Mechanism Mk 26 Mod 0, LD 497745 shall operate within 7 seconds maximum when exposed to a constant acceleration of 7g plus or minus 1g.

3.3.3 Parachute deployment. The parachute, shroud lines, and steel cable shall be free of any twist or kinks and folded as specified in OD 13799.

3.4 Environmental requirements. Unless otherwise specified, the pack assembly shall be capable of conforming to the performance and product characteristics of this specification after being exposed to the environments and all conditions specified herein.

3.4.1 Humidity. The pack assembly shall withstand relative humidity up to 95 percent, as specified in MIL-E-5272, Procedure I.

3.4.2 Vibration. The pack assembly shall show no evidence of separation of the cover assembly from the body when subjected to vibration requirements specified in 4.5.1.2.

3.5 Workmanship. The workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with the requirements of this specification. All parts used in the assembly shall be thoroughly clean and free of excess materials, chips, and loose or splattered foreign materials. The standards of workmanship exhibited in any approved preproduction sample, subject to any quqlification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. Inspection of pack assembly shall be classified as follows:

a. Preproduction and periodic production inspection (see 4.5).

b. Quality conformance inspection (see 4.6).

4.3 Sampling. (See 6.2.1.)

4.3.1 Lot size. The term "lot" shall mean "inspection lot", that is, a collection of product submitted by a supplier for inspection. Unless otherwise specified in the contract or order, the number of product in each lot shall be one hundred and fifty.

4.3.2 Preproduction sample. A preproduction sample of five pack assemblies manufactured in accordance with 3.2, shall be subjected to the preproduction tests detailed in 4.5. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the pack assembly by the supplier, prior to approval of the preproduction sample, shall be at the supplier' s risk.

4.3.3 Periodic production sample. From each succeeding lot of three hundred or less units that have satisfactorily passed the quality conformance inspection of 4.6, a random sample of two pack assemblies shall be selected. These sample units shall be subjected to the tests of 4.5. The pack assemblies shall be considered satisfactory if they meet all specifications and pertinent drawing requirements and withstand the tests specified in 4.5. (See Table 1.) When a defective unit is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

a. Determine the cause of the defect.

b. Take the necessary action with the concurrence of the procuring activity, to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

c. Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

4.3.4 Quality conformance inspection sampling. Samples shall be selected as specified in 4.6 and shall be used to determine the acceptability of the pack assemblies inspected as specified in 4.6.

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4.4 Test equipment. The following items of test equipment are required to perform tests herein:

Equipment	Capability
a. Centrifuge	0-8g; maintain constant acceleration within plus or minus 1g. Arm length to be 45 inches minimum. Test item including test fixtures weights approximately 30 pounds.
b. Air Cylinder	Capable of delivering 80 pounds of force.

4.5 Preproduction and periodic production inspections. The preproduction and periodic production samples (Table I), shall be subjected to the inspections herein.

4.5.1 Test procedure. The pack assembly shall be subjected to the tests of Table I as outlined herein. Unless otherwise specified, inability of the pack assemblies to pass the quality conformance inspection tests, subsequent to the environmental tests, constitute a failure.

Table I. Preproduction and Periodic Production Test Sequence

Test	Paragraph Number	Test Quantities Preproduction	Periodic
Humidity	4.5.1.1	5	-
Vibration	4.5.1.2	2	2
Quality Conformance inspection Test	4.6.2	5	2

4.5.1.1 Humidity. Unless otherwsie specified in the contract or order, this test shall be performed on preproduction sample units only. The pack assembly shall be subjected to Humidity Test, Procedure I as specified in MIL-E-5272.

4.5.1.2 Vibration. The pack assembly shall be restrained at the circumference of the body, at the midpoint, and vibrated in each of the three axes, Figure 1, at the maximum resonance frequency, and at a frequency range of 10 to 60 cps at an amplitude not to exceed 1.3g for a period of thirty minutes.

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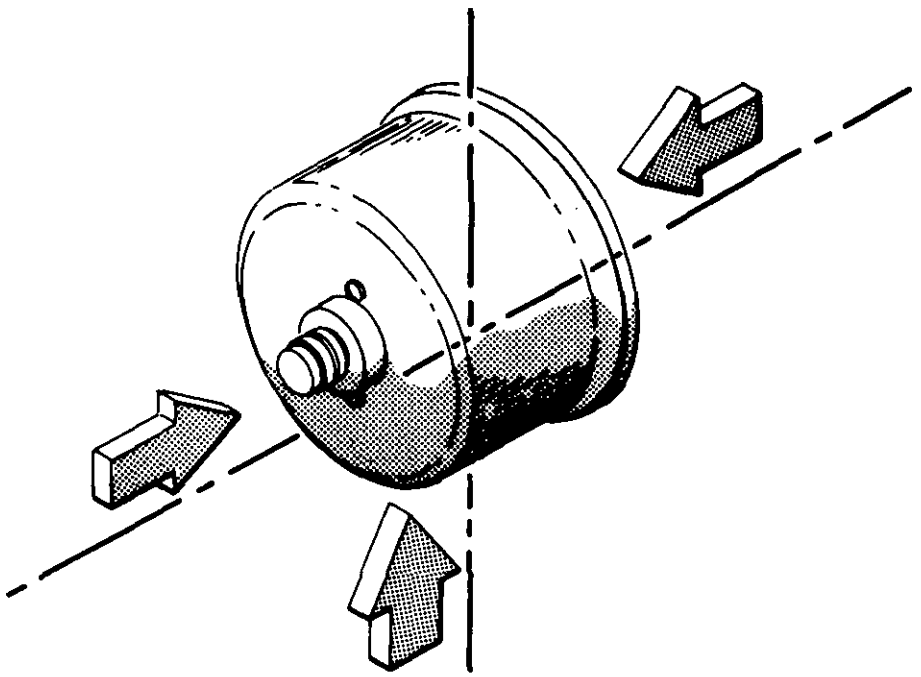


Figure 1. Vibration Axes.

4.6 Quality Conformance Inspection. Samples selected using the plans and procedures set forth in Table II of this specification, shall be subjected to the test and examinations as described in 4.6.2.

Table II. Quality Conformance Tests and Examinations and AQL

Test or Examination	Sampling Plan	AQL	Test Paragraph
Non-Operative	*	2.5	4.6.2.1
Operative	*	2.5	4.6.2.2
Deployment	*	2.5	4.6.2.3
Visual examination	**	**	4.6.2.4
Preservation, packaging packing and marking	**	**	4.7

* MIL-STD-105D Special Inspection Level S-3; zero defects accepted.
** One-hundred percent inspection; defect is cause for rejection of individual unit.

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4.6.1 Test conditions. *Unless otherwise specified,* the pack assembly shall be subjected to the qualify conformance inspection under the following conditions.

a. Temperature - Room ambient (65 to 95 degrees F)

b. Altitude - Normal ground atmospheric pressure

c. Humidity (Relative) - Room ambient up to 95 percent

4.6.2 Test procedure. The release mechanism or pack assembly shall be subjected to the tests and examinations as outlined herein. Unless otherwise specified, inability to pass the tests and examinations shall constitute a failure.

4.6.2.1 Nonoperative. Mount the release mechanism as shown in Figure 2, and attach test weight, Figure 3, to shaft nut. Subject assembly to an acceleration of 4 plus 0, minus 1g. Actuate air cylinder for pin withdrawal. Nut release shall not occur. Remove unit and reassemble.

4.6.2.2 Operative. The release mechanism shall be mounted as specified in 4.6.2.1. Subject assembly to an acceleration of 7 plus or minus 1g. Actuate air cylinder and record time of pin withdrawal. Nut release shall occur and the elapsed time between pin withdrawal and release shall be 7 seconds maximum.

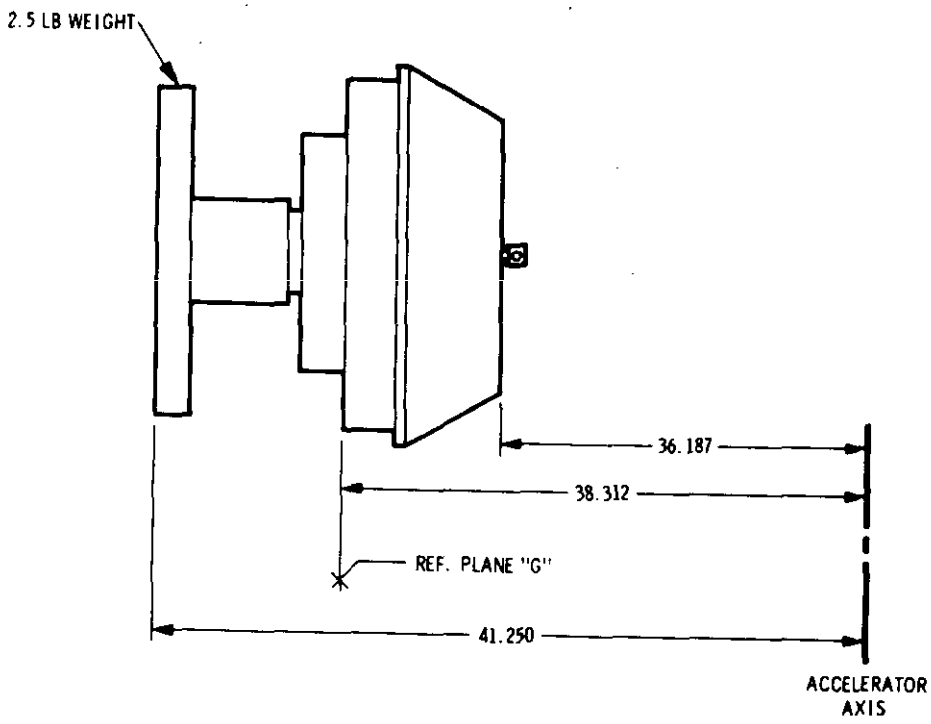
NOTE: Pin release and nut release may be recorded as shown in Figure 4, details A and B.

4.6.2.3 Parachute deployment. Remove shield and uncoil steel cable to full length with both ends rotationally secured and inspect for proper coiling procedures and that the cable is free of twists or kinks. Remove parachute from container, untie bowline knot securing pin lanyard to suspension lines at center vent and extend parachute to full length. Unfold parachute and verify correct folding procedures as specified in OD 13799. Inspect suspension lines for twisting and verify that suspension lines are not tangled with the nylon cords. Repackage in accordance with OD 13799.

4.6.2.4 Visual examination. Visual examination shall be made on each pack assembly submitted for delivery to the procuring agency. The pack assembly shall be examined for conformance to the requirements of good workmanship in 3.5. Nonconformance is cause for rejection of the individual unit.

4.7 Preservation, packaging, packing, and marking. The pack assembly shall be examined for conformance to the prescribed preservation, packaging, packing, and marking requirements of Section 5.

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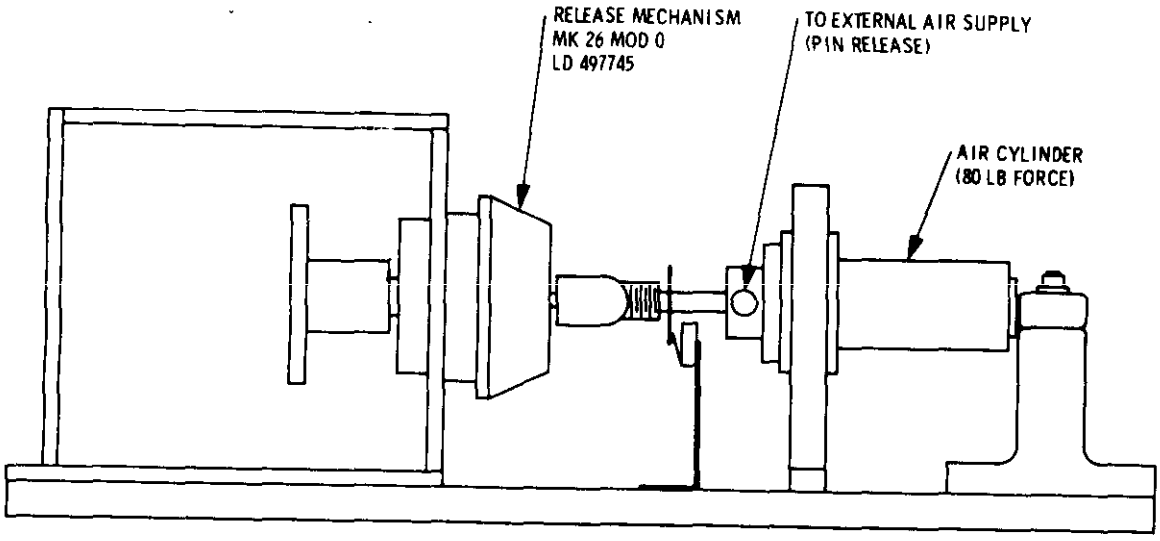


Figure 2. Suggested Functional Test Fixture

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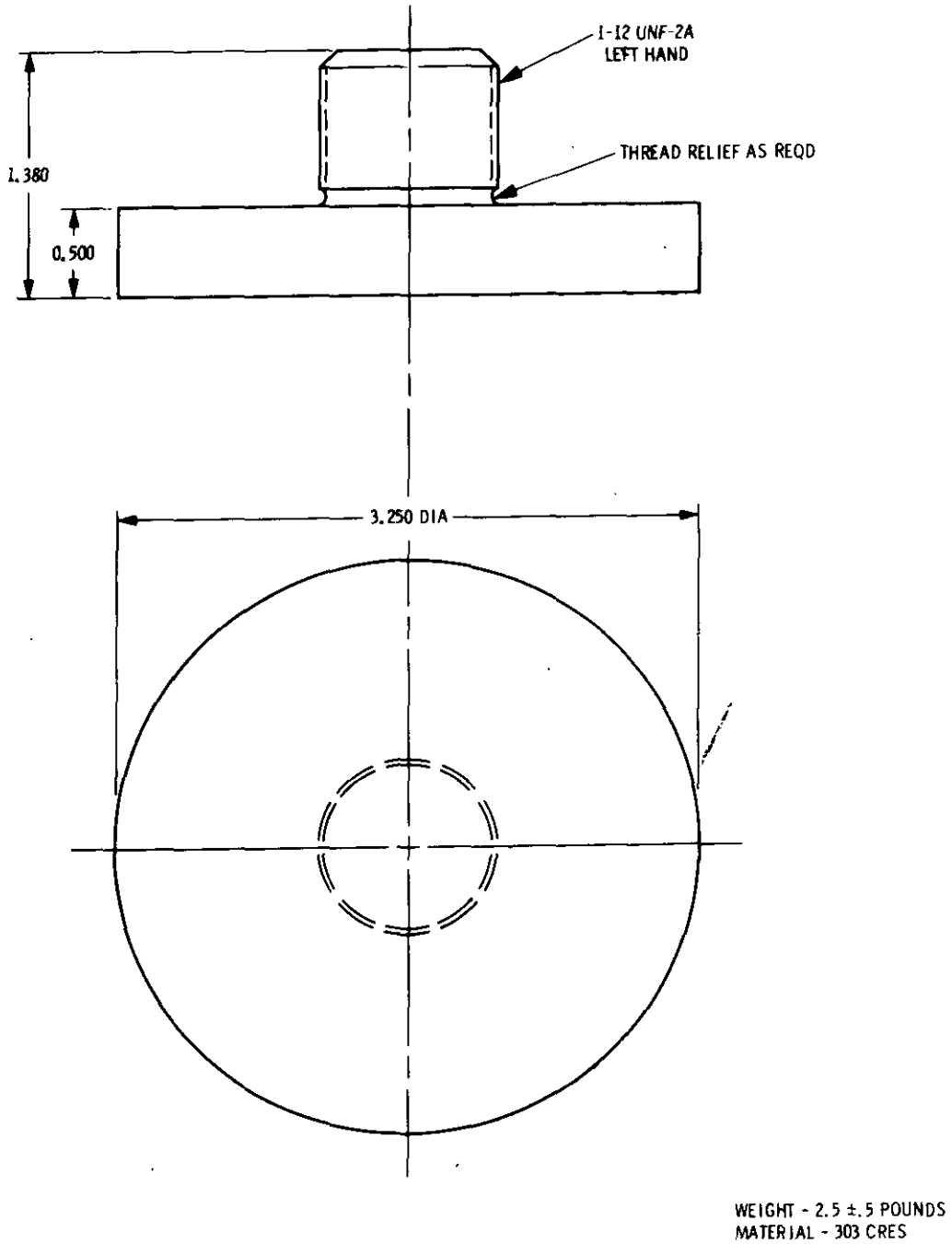
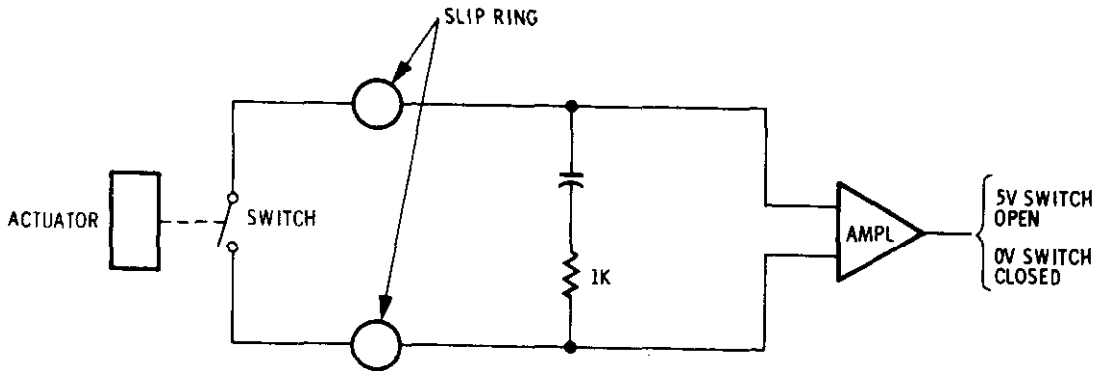
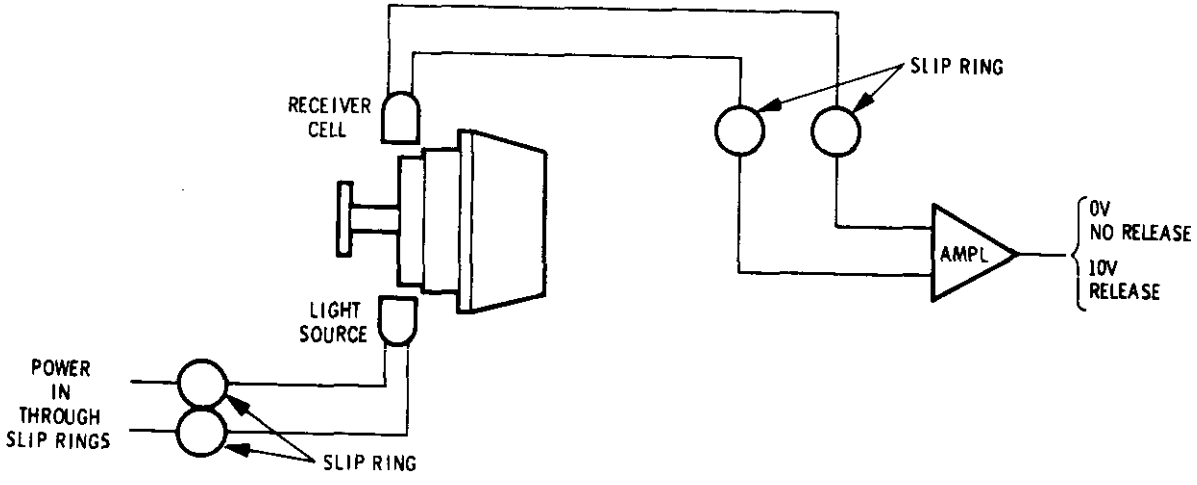


Figure 3. Test Weight

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Detail A



Detail B

Figure 4. Suggested Method of Recording Pin Release and Nut Release

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A.

5.1.1 Level A. Preservation and packaging shall be in accordance with LD 497770.

5.1.1.1 Cleaning. The pack assembly shall be cleaned in accordance with Method C-1, Specification MIL-P-116.

5.1.1.2 Drying. The pack assembly shall be dried in accordance with Method D-1 or D-4, Specification MIL-P-116.

5.2 Packing. Packing shall be Level A.

5.2.1 Level A. Unless otherwise specified in the contract, order or requisition the pack assembly shall be packed for shipment in accordance with LD 497771.

5.3 Marking. In addition to any special marking required by the contract, or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use and description.

6.1.1 Intended use. The pack assembly is intended to be used for missile stabilization and deceleration.

6.1.2 Description. The pack assembly is a device designed for parachute deployment and to separate from missile at water entry.

6.2 Ordering data. Procurement documents should specify the following:

a. Title number and date of this specification.

b. Any exceptions to this specification.

6.2.1 Additional samples. Additional samples required because of failure of any sample(s) to pass the prescribed inspections shall be provided and inspected at the expense of the supplier.

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